Aditya Dinesh Rajput

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Embedded Systems | Linux | Robotics | Web Development

Education

PDPM IIITDM Jabalpur — CGPA: 9.2
B.Tech in Electronics and Communication Engineering — *2024 – 2028*

Technical Skills

Languages: C, C++, Python, Rust, Java, JavaScript, Lua
Systems & Embedded: Linux, STM32, UART, SPI, I2C, RTOS, ARM Assembly
Tools: Git, GDB, QEMU, STM32CubeIDE, NeoVim, MATLAB
Frameworks: TensorFlow, PyTorch, React

Projects

Real Time Operating System (RTOS) | C, RTOS, ARM Assembly — GitHub Repo

- Implemented a small RTOS from scratch in C.
- It uses Preemptive Scheduling for scheduling tasks.
- Implemented the context-switcher in ARM Assembly.
- It has many features like Queues, Semaphores, and Mutexes.
- Debugged and tested using GDB.

STM32F411 Bare-Metal Device Drivers | Embedded C, STM32CubeIDE — GitHub Repo

- Implemented bare-metal register-level drivers for GPIO, UART, SPI, and I2C drivers for STM32F411 without using HAL libraries.
- Configured peripheral clocks, memory-mapped registers, and communication protocols directly using the STM32 reference manual.
- Debugged baud-rate and peripheral clock issues during UART driver development.

Self Balancing Bot Simulation | CoppeliaSim, Python

- Designed and simulated a self-balancing robot in CoppeliaSim using PID-based stabilization and real-time sensor feedback.
- Fine-tuned PID controller parameters to improve balance stability and response time.

VCG Signal Compression Using Machine Learning | TensorFlow, Signal Processing — GitHub Repo

- Developed a CNN-LSTM autoencoder model for VCG signal compression and reconstruction.
- Trained and evaluated the model using TensorFlow-based workflows for biomedical signal processing.

Chess Vision System | Python, OpenCV, YOLO, Stockfish — GitHub Repo

- Developed a computer vision pipeline that detects chess pieces and reconstructs board state using YOLO-based object detection.
- Generated FEN strings from detected board positions and integrated Stockfish to suggest optimal moves.
- Implemented board warping and grid mapping using OpenCV for accurate position tracking.

Experience

E-Yantra 2025 — IIT Bombay
Reached Simulation Stage

- Developed a simulation for a self-balancing robot in CoppeliaSim.

Techkriti 2026 - IARC Competition — IIT Kanpur
1st Place

- Developed and optimized an autonomous maze-solving robot that achieved the fastest completion time in the competition.

Leadership & Extracurricular

Electronics and Robotics Society — IIITDM Jabalpur
Core Committee Member

- Organized workshops and events focused on embedded systems, robotics, and electronics for students.
- Mentored first-year students during RoboRush 4.0 in building WiFi-controlled robots.